

Laboratory session 1

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25. September 2026

1 Laboratory session 1

The first experiment is using standing waves inside a long tube in order to measure the speed of sound inside the gas the tube is filled with. The tube and experiment settings are specified by the following data:

Tube length: $L = 1244 \pm 1.5 \text{ mm}$.

input signal: $V_0 = 35 \text{ mV}$

Tubes:

K1: argon or CO_2 at $T = T_{\text{room}}$

K2: air at $T = T_{\text{room}}$

K3: air at $T = 70^\circ\text{C} = 343.15 \text{ K}$

K4: air at $T = 50^\circ\text{C} = 323.15$

$T_{\text{room}} = 21^\circ\text{C} = 294.15$

We analyzed the properties of the gas in tube K4. The resistance of the tube is measured to be $R = (34.55 \pm 0.01) \text{ k}\Omega$.

2 Tasks

2.1 Measurement of ν_k

We measured the following resonance frequencies ν_k between 200 Hz and 2 kHz :

K4: $\nu_k = [313 \pm 1, 450 \pm 1, 594 \pm 1, 740 \pm 1, 886 \pm 1, 1033 \pm 1, 1178 \pm 1, 1326 \pm 1, 1473 \pm 1, 1620 \pm 1, 1767 \pm 1, 1913 \pm 1]$

The other groups measured:

K3: $\nu_k = [301 \pm 1, 459 \pm 1, 607 \pm 1, 757 \pm 1, 907 \pm 1, 1057 \pm 1, 1208 \pm 1, 1359 \pm 1, 1510 \pm 1, 1661 \pm 1, 1813 \pm 1, 1963 \pm 1]$

K2: $\nu_k = [286 \pm 1, 419 \pm 1, 470 \pm 1, 563 \pm 1, 700 \pm 1, 840 \pm 1, 979 \pm 1, 1117 \pm 1, 1258 \pm 1, 1397 \pm 1, 1537 \pm 1, 1677 \pm 1, 1816 \pm 1, 1955 \pm 1]$

K1: argon: $\nu_k = [265 \pm 1, 393 \pm 1, 521 \pm 1, 650 \pm 1, 778 \pm 1, 905 \pm 1, 1035 \pm 1, 1164 \pm 1, 1293 \pm 1, 1423 \pm 1, 1551 \pm 1, 1680 \pm 1, 1810 \pm 1, 1938 \pm 1]$

CO_2 : $\nu_k = [225 \pm 1, 440 \pm 1, 544 \pm 1, 656 \pm 1, 764 \pm 1, 872 \pm 1, 979 \pm 1, 1091 \pm 1, 1199 \pm 1, 1209 \pm 1, 1418 \pm 1, 1527 \pm 1, 1635 \pm 1, 1745 \pm 1, 1854 \pm 1, 1963 \pm 1]$

2.2 Estimated error

We estimated our error to be around 1 Hz

2.3 The best linear fit

In order to find the speed of sound in the gas, we want to use the equation $\nu_n = an + b$. However, we do not know which resonance frequency we are measuring, so we do not know which n belongs to which measured ν_n . This is why we set the reference of the measured ν_n to the first one measured. Therefore we get:

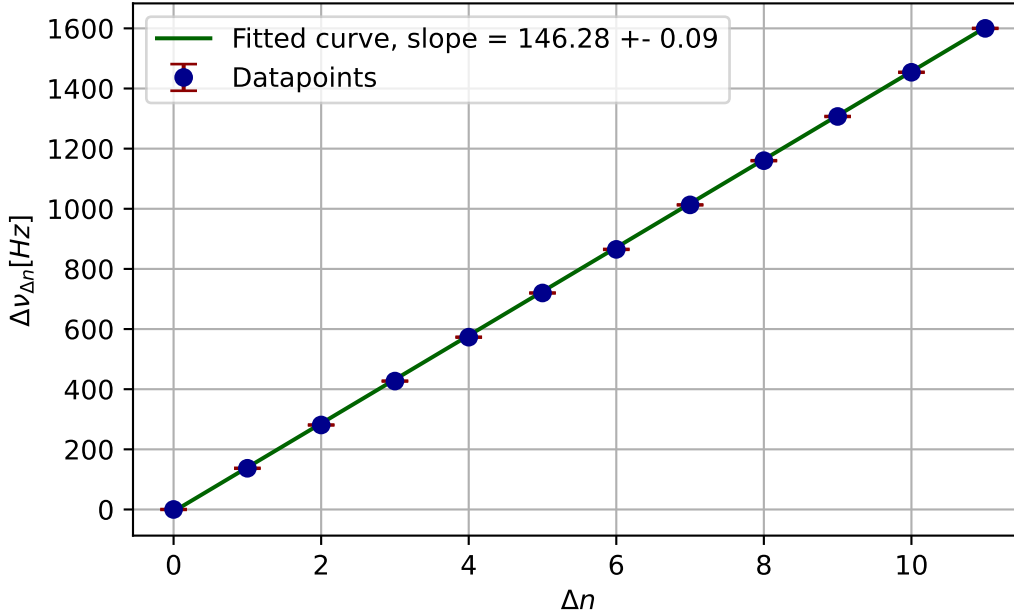
$$\nu_n - \nu_{n_0} = \Delta\nu_{\Delta n} = a\Delta n + b - (a(n_0) + b) = a(n - n_0) + b - b = a\Delta n = \frac{c}{2L}\Delta n$$

So the central relation is:

$$\Delta\nu_{\Delta n} = \frac{c}{2L}\Delta n$$

where Δn starts at 0 for the first measurement and for k measurements continues until $k - 1$ for the k^{th} resonance frequency. $\Delta\nu_{\Delta n}$ against Δn is plotted and evaluated in the following:

K4:

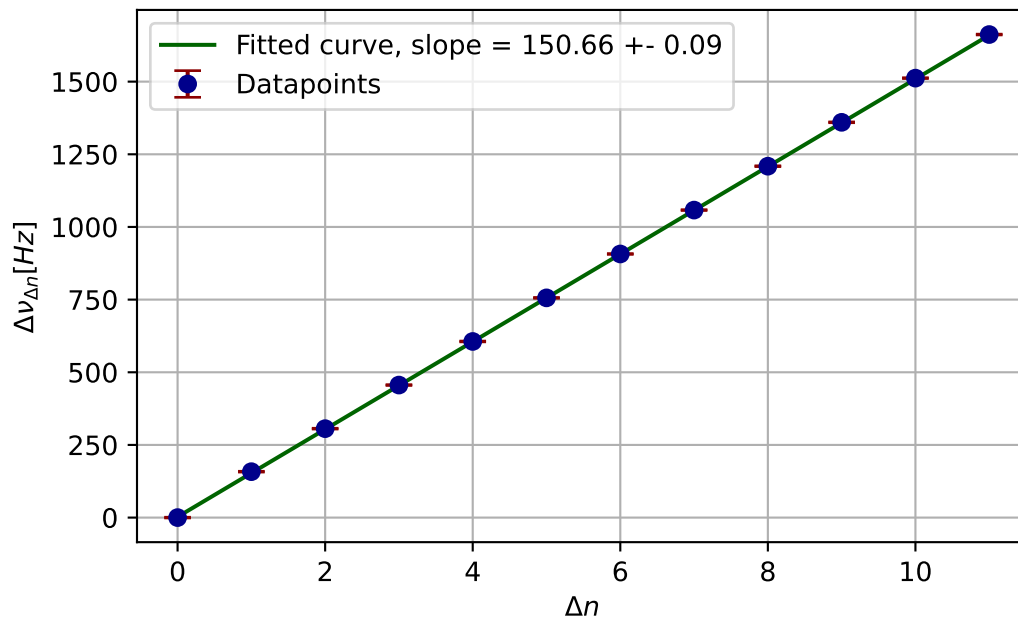


Given the slope of the fit curve, the velocity of sound can be calculated:

$$\text{speed of sound} = 363.95 \pm 0.49 \text{ m/s}$$

K3:

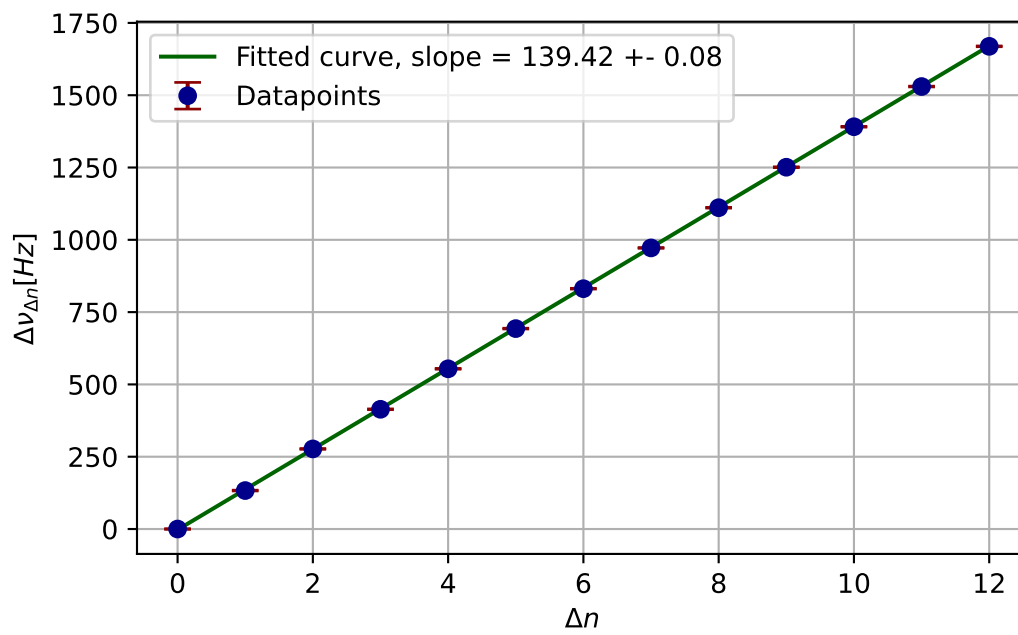
$$\text{The resistance of the tube is measured to be } R = (15.66 \pm 0.02)k\Omega$$



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 374.85 ± 0.50 m/s

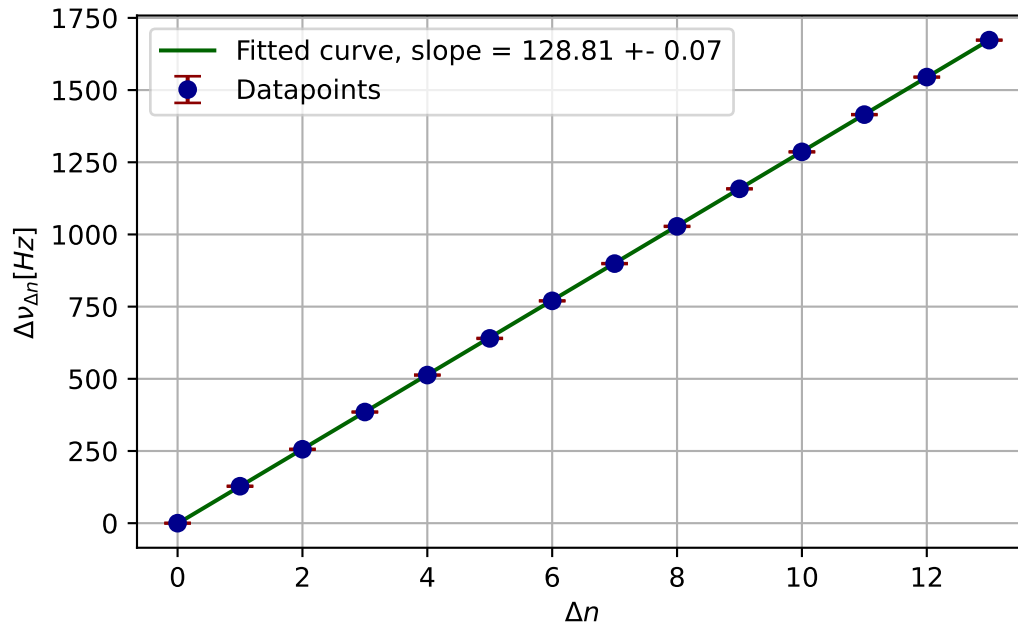
K2:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 346.87 ± 0.46 m/s

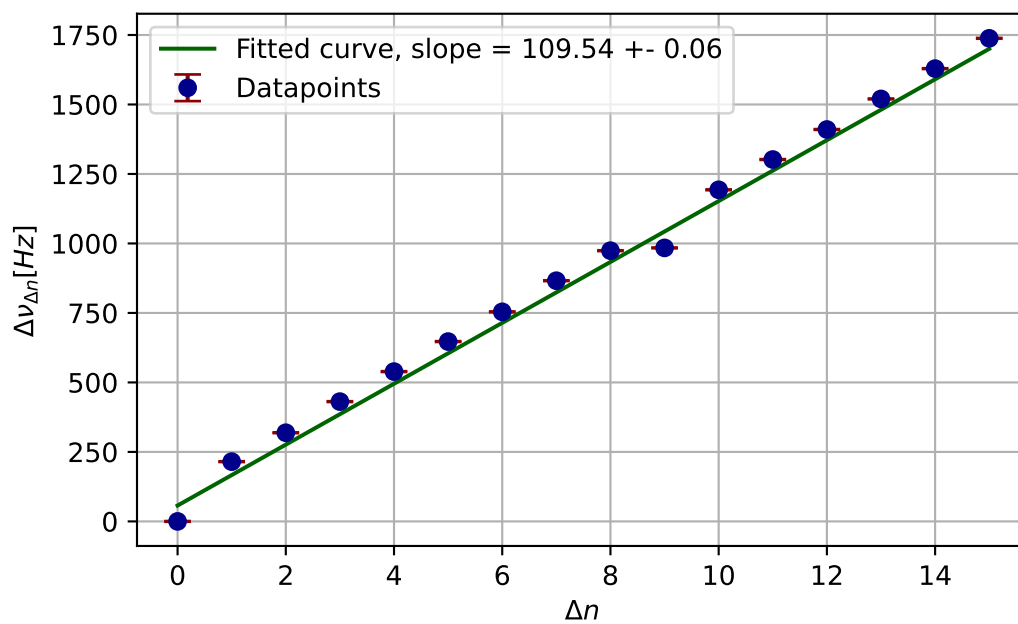
K1, argon:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 320.47 ± 0.42 m/s

K1, CO_2 :



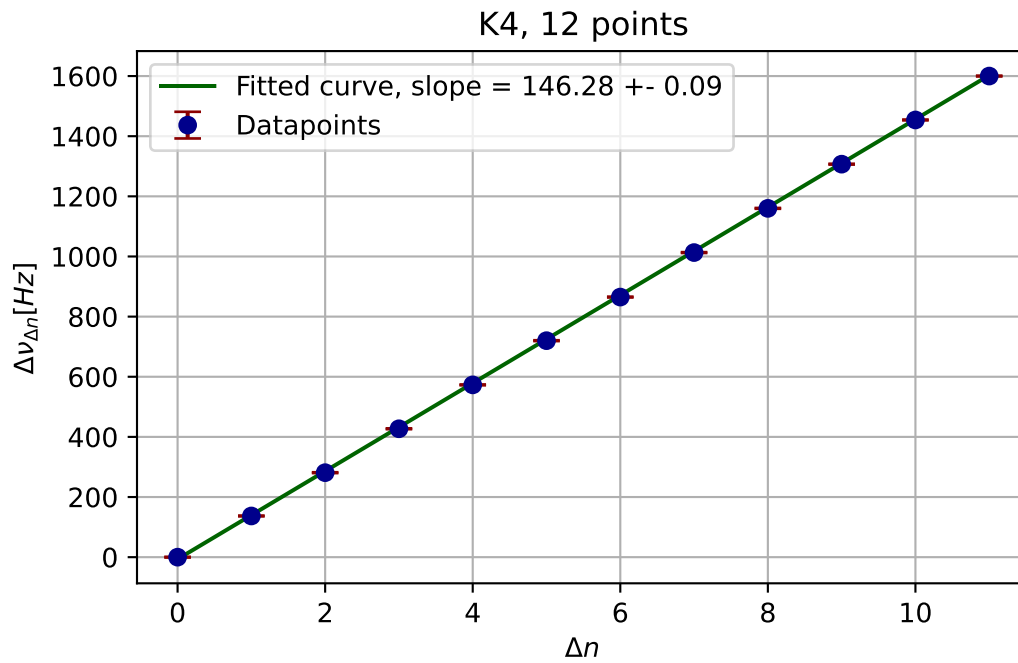
Given the slope of the fit curve, the velocity of sound can be calculated:

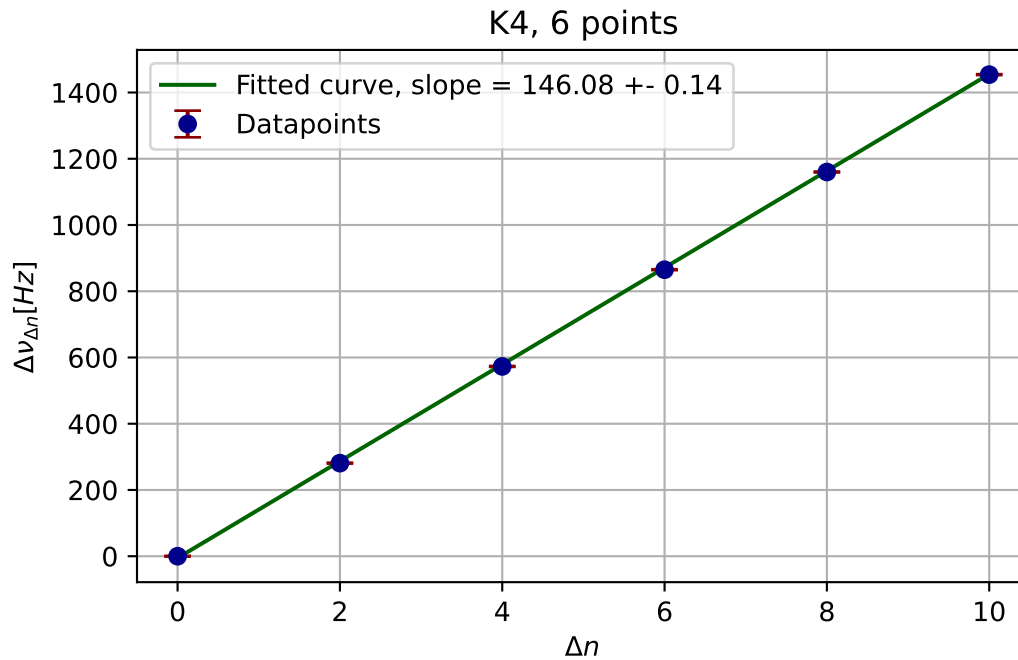
speed of sound = 272.53 ± 0.36 m/s

2.4 Most important contribution to uncertainty

The most important contribution to the error is the inaccuracy in finetuning the resonance frequency. We estimated our accuracy in doing so to $1Hz$.

The uncertainty shrinks the more points are included in the fit. In the following we see the plot for K4, once with 12 points and once with 6:





As we can see, the uncertainty on the slope is more than 50% greater if we only choose half of the points.

2.5 Degrees of freedom

For an ideal gas, the velocity of sound is given by

$$c(T) = \sqrt{\frac{(f + 2)RT}{fM_{mol}}}$$

where f are the degrees of freedom, R is the gas constant, T the temperature of the gas and M_{mol} the molar mass. This equation can be solved for f :

$$f = \frac{2RT}{c^2 M_{mol} - RT}$$

We calculate for f :

K4:

The calculated amount of degrees of freedom from the experiment is 5.81 ± 0.09

K3:

The calculated amount of degrees of freedom from the experiment is 3.89 ± 0.05

K2:

The calculated amount of degrees of freedom from the experiment is 4.71 ± 0.07

K1, argon:

The calculated amount of degrees of freedom from the experiment is 2.94 ± 0.03

K1, co2:

The calculated amount of degrees of freedom from the experiment is 5.95 ± 0.10

In theory, the gases inside K?,... should have ? degrees of freedom. These consist of ...

2.6 Comparison to simulation and theory

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2.7 Phase diagram

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2.8 C_V , Z and degrees of freedom of a simulated lennard jones gas

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2.9 Different temperatures at fixed gas density

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2.10 Temperatures above the critical point

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2.11 Determination of C_V

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